



Technical University of Denmark (DTU)

Department of Applied Mathematics and Computer Science (DTU Compute)

MASTER'S THESIS PROJECT PROPOSAL

**Explainable Self-Supervised Network Anomaly
Detection for Cyber Defense Using Multi-Modal
Network Telemetry**

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1 Introduction

Cyber threats such as zero-day attacks, advanced persistent threats (APTs), and insider threats are very difficult to detect using traditional signature-based intrusion detection approaches. Self-supervised learning (SSL) offers a promising approach by learning meaningful patterns from large amounts of unlabeled network telemetry, reducing the dependence on labeled attack data. Many AI-based anomaly detection systems lack transparency, restricting their usefulness in operational cybersecurity environments. This thesis proposes an explainable self-supervised network anomaly detection framework that uses multi-modal network telemetry to identify unknown cyber threats while providing interpretable explanations to support cybersecurity analysts.

The proposed framework is designed to operate on network telemetry collected from security monitoring platforms such as Suricata and Zeek, which generate flow, DNS, TLS, HTTP, protocol, and alert logs. These telemetry sources provide complementary information about network behavior and form the basis for learning normal communication patterns in operational cyber defense environments such as Terma.

2 Background

The increasing complexity of modern cyber threats has made traditional signature-based intrusion detection systems insufficient for identifying previously unseen attacks. Security monitoring platforms such as Suricata and Zeek continuously generate large volumes of network telemetry, including network flows, DNS, TLS, HTTP, and protocol logs. While these data sources provide valuable information about network activities, identifying malicious behaviour within them remains challenging, particularly due to the limited availability of labeled cybersecurity datasets.

Self-supervised learning has recently appeared as a promising approach for learning meaningful representations from unlabeled data, making it well suited for anomaly detection in cybersecurity. However, many existing machine learning models operate as black boxes, limiting their adoption in operational environments where security analysts require clear and trustworthy explanations for generated alerts. Combining self-supervised learning with explainable artificial intelligence (XAI) and multi-modal network telemetry offers the potential to improve both detection performance and analyst confidence.

3 Problem Statement

Current network intrusion detection systems primarily rely on predefined signatures or rule-based methods, supervised learning methods that require labeled attack data. These approaches are often ineffective against novel or developing threats and provide limited insight into why an event is classified as anomalous. There is therefore a need for an anomaly detection framework capable

of learning normal network behavior from unlabeled multi-modal telemetry while producing interpretable explanations for security analysts.

How can self-supervised learning and explainable AI be combined to detect previously unseen cyber threats from network telemetry while providing interpretable decision support for cybersecurity analysts?

4 Research Questions

- **RQ1:** How effectively can self-supervised learning detect previously unseen network attacks without labeled training data?
- **RQ2:** How does combining multiple telemetry sources improve anomaly detection performance?
- **RQ3:** Which Explainable AI methods provide meaningful explanations for anomaly detection decisions?
- **RQ4:** What trade-offs exist between detection performance and false positive reduction in SSL-based anomaly detection?
- **RQ5:** How suitable is the proposed approach for deployment within Terma's cyber defense environments?

5 Research Objectives

The primary objective is to design, implement, and evaluate an explainable self-supervised anomaly detection framework capable of identifying unknown cyber attacks from multi-modal network telemetry. Specific objectives include:

1. Study existing anomaly detection techniques
2. Review self-supervised learning methods for cybersecurity
3. Design a multi-modal telemetry representation
4. Develop role-aware behavioral baselines for different asset types, network zones, or device roles to improve anomaly detection accuracy.
5. Develop self-supervised representation learning
6. Detect anomalies using learned latent embeddings
7. Integrate Explainable AI techniques
8. Evaluate against benchmark cybersecurity datasets

9. Compare with supervised and unsupervised baselines
10. Analyze deployment feasibility for defense environments

6 Methodology

This research follows a design and evaluation methodology to develop an explainable self-supervised anomaly detection framework for cyber defense. The study will combine literature review, dataset preparation, data preprocessing, model development, and explainability analysis. The overall goal is to learn normal network behavior from unlabeled multi-modal telemetry and detect previously unknown cyber threats while providing interpretable explanations for security analysts.

6.1 Literature Review

A systematic literature review will be conducted on research on network anomaly detection, self-supervised learning (SSL), and explainable artificial intelligence (XAI) in cybersecurity. The literature review will focus on recent studies published in major journals, conference proceedings, and industry reports. Particular attention will be given to SSL methods that learn representations from unlabeled network data and XAI techniques that improve analyst trust in machine learning systems. The findings from the literature review will be used to identify research gaps, justify the proposed approach, and guide the selection of algorithms and evaluation methods.

6.2 Dataset

The framework will be evaluated using publicly available cybersecurity datasets that contain network traffic and protocol logs. Possible datasets include CSE-CIC-IDS2018, CIC-DDoS2019, UNSW-NB15, CIC-IDS2017, and other benchmark intrusion detection datasets that provide flow-based and application-level telemetry. If access is available, anonymized telemetry from Terma A/S may also be used for additional validation. The selected datasets will include multiple telemetry sources such as network flows, DNS records, HTTP logs, and TLS information to support multi-modal analysis. Using public benchmark datasets to ensure that the results can be reproduced and compared with existing research.

The network telemetry used in this research will primarily consist of data generated by monitoring systems such as Suricata and Zeek, including network flows, DNS, TLS, HTTP, protocol metadata, and security alerts. If access is granted, anonymized telemetry collected from Terma's operational environment will be used to validate the proposed framework.

6.3 Data Preprocessing

Before model training, the raw network telemetry will be cleaned and transformed into a consistent format. Missing values, duplicate records, and corrupted entries will be removed. Numerical features will be normalized, and categorical fields such as protocol types and service names will be encoded using suitable encoding techniques. Telemetry from different sources will be synchronized using timestamps and merged into a unified representation. Where metadata is available, network assets will be grouped according to their functional roles, network zones, or device categories. This role-aware organization enables the framework to learn separate normal behavioral profiles for different classes of assets, improving the detection of context-specific anomalies. Feature engineering will be applied to create meaningful aspect, including packet data, connection duration, protocol behavior, and traffic patterns. The processed data will then divided into training, validation, and testing sets.

6.4 Model Development

The proposed framework will use self-supervised learning to learn latent representations of normal network behavior from unlabeled data. Instead of learning a single global behavioral model, the proposed framework will also investigate role-aware anomaly detection by learning separate behavioral representations for different asset types, network zones, or host roles whenever sufficient metadata is available. This approach is expected to reduce false positives and improve the prioritization of suspicious activities. Initial experiments will begin with baseline models such as autoencoders and contrastive learning methods. More advanced SSL architectures will then be explored to improve representation quality. The learned establishings will be used by an anomaly detection component that identifies traffic patterns that deviate significantly from normal behavior.

The performance of the proposed model will be compared with traditional supervised and unsupervised anomaly detection approaches using metrics such as accuracy, precision, recall, F1-score, ROC-AUC, false positive rate, PR-AUC, detection latency, and Matthews correlation coefficient (MCC). Whenever possible, experiments will follow a time separated evaluation strategy, where models are trained using historical network telemetry and evaluated on later observations. This evaluation better reflects real world deployment scenarios and measures the model ability to detect previously unseen anomalies.

6.5 Explainable AI

Explainability techniques will be integrated into the anomaly detection framework to provide understandable reasons for each alert. Methods such as SHAP, LIME, and feature attribution analysis will be evaluated to identify which telemetry features contribute most strongly to anomaly decisions. The generated explanations will be presented in a form that is useful for

cybersecurity analysts, including ranked feature importance and behavioral summaries. The quality of explanations will be assessed based on clarity, consistency, and usefulness for incident investigation. This step aims to increase analyst trust and support the practical deployment of the proposed system in operational cyber defense environments.

7 Expected Outcomes

The expected outcome of this research is an explainable anomaly detection framework capable of identifying previously unknown cyberattacks using unlabeled network telemetry. The framework is expected to reduce false positives while providing meaningful explanations that improve analyst trust and support faster incident investigation. The results will contribute to the integration of trustworthy artificial intelligence into real-world cyber defense system and provide valuable insights for future deployment within industrial environments such as Terma A/S.

8 Expected Contributions

8.1 Academic Contributions

- Proposed self-supervised framework for network anomaly detection
- Multi-modal telemetry representation
- Explainable AI integration
- Comprehensive benchmark evaluation

8.2 Industrial Contributions

- Reduced dependence on labeled attack data
- Improved detection of zero-day attacks
- Enhanced analyst trust through explainability
- Prototype suitable for future integration into Terma’s cyber defense platforms
- Role-aware anomaly prioritization that improves the usefulness of alerts for cybersecurity analysts operating Terma’s security monitoring systems.

9 Conclusion

This thesis aims to develop and evaluate an explainable self-supervised anomaly detection framework for cyber defense. By combining SSL, multi-modal telemetry fusion, anomaly detection

techniques, and explainable AI methods, the proposed system search aims to improve the detection of previously unseen attacks while reducing false positives and increasing analyst trust. The proposed framework is intended to complement existing signature-based detection systems, such as Suricata and Zeek, by providing an additional machine learning based anomaly detection capability rather than replacing existing security monitoring mechanisms. The expected outcome is a practical and explainable anomaly detection prototype that contributes to both academic research and industrial cybersecurity applications, particularly in mission critical environments such as Terma A/S.

10 Risk and Mitigation

Table 1: Risk and Mitigation

Risk	Mitigation
Limited access to industrial data	Use public benchmark datasets and validate the methodology on anonymized Terma data, if available.
High false positives	Apply hyperparameter tuning, threshold optimization, and ensemble methods to improve detection performance.
Model complexity	Begin with baseline architectures before exploring advanced self-supervised learning (SSL) models.
Explainability overhead	Evaluate lightweight explainability methods and optimize them for practical deployment.

Table 2: Project Timeline

Duration	Academic Activities	Industrial (Terma) Activities
Month 1	Conduct a literature review on self-supervised learning, network anomaly detection, and Explainable AI. Finalize the research problem, objectives, and methodology.	Understand Terma’s IDS environment, security monitoring workflow, and available telemetry sources (e.g., Suricata and Zeek). Define the project scope and expected outcomes with the company supervisor.
Month 2	Collect and preprocess datasets, perform feature engineering, and establish baseline anomaly detection models.	Study Terma’s network telemetry structure and prepare data pipelines using available or anonymized telemetry. Discuss data availability and validation strategy with the company supervisor.
Month 3	Design, implement, and train the proposed self-supervised anomaly detection framework. Optimize the model through iterative experiments.	Validate the proposed approach against Terma’s operational requirements, incorporate company feedback, and investigate role-aware behavioural modelling where applicable.
Month 4	Integrate Explainable AI techniques such as SHAP and LIME, and evaluate the framework against baseline approaches using standard performance metrics.	Assess whether the generated alerts are understandable and useful for cybersecurity analysts. Review intermediate results with the company supervisor and refine the framework based on feedback.
Month 5	Perform comprehensive evaluation, analyse the results, discuss limitations, and write the thesis.	Evaluate the deployment feasibility of the proposed framework within Terma’s IDS environment and prepare recommendations for future industrial integration.
Month 6	Revise the thesis based on supervisor feedback, finalize the document, prepare the presentation, and defend the thesis.	Present the final prototype and research findings to Terma, discuss future improvements, and document potential directions for industrial deployment.